

Lab 03: Real-World Data Linear Regression

Topic: The Mental Health of People — National Suicide Mortality Rates by Sex

Part A — Data

1. Source

World Health Organization (WHO), Global Health Observatory (GHO) — "Age-standardized suicide rates (per 100,000 population)," 2021 estimates.

URL: [https://www.who.int/data/gho/data/indicators/indicator-details/GHO/age-standardized-suicide-rates-\(per-100-000-population\)](https://www.who.int/data/gho/data/indicators/indicator-details/GHO/age-standardized-suicide-rates-(per-100-000-population))

Values were taken from the WHO 2021 country table as compiled and reproduced in the Wikipedia article "List of countries by suicide rate": https://en.wikipedia.org/wiki/List_of_countries_by_suicide_rate

2. Description of the data

Suicide is classified by the WHO as a major global mental-health outcome: it is one of the clearest, most consistently measured population-level indicators of a country's mental-health burden. The WHO GHO publishes the age-standardized suicide mortality rate (deaths per 100,000 population per year) for each of its member states, broken down by sex. This lab uses the male and female age-standardized rates for 20 countries, drawn deliberately from a wide range of regions and income levels (Africa, Asia, Europe, and the Americas; high-, upper-middle- and lower-middle-income countries) so that the sample is not dominated by any single region.

3. Variables

Independent variable, x : Male suicide mortality rate, age-standardized, per 100,000 male population (2021).

Dependent variable, y : Female suicide mortality rate, age-standardized, per 100,000 female population (2021).

Both variables are reported by WHO for the same country and year, so pairing them lets us ask a mental-health question directly: do countries with a higher male suicide rate also tend to have a higher female suicide rate, and by how much?

4. Table of observations

#	Country	Male rate x (per 100,000)	Female rate y (per 100,000)
1	Egypt	0.9	0.3
2	Saudi Arabia	1.1	0.7
3	Turkey	4.3	1.1
4	Bangladesh	3.6	2.0
5	Mexico	11.6	2.6
6	Italy	10.5	3.6
7	Brazil	11.4	3.9
8	Spain	13.1	4.5
9	United Kingdom	14.7	4.6
10	Canada	14.2	4.7

#	Country	Male rate x (per 100,000)	Female rate y (per 100,000)
11	Netherlands	15.4	7.6
12	Germany	18.4	7.6
13	Australia	19.5	6.7
14	United States	24.7	6.5
15	Sweden	18.9	8.7
16	Japan	19.6	9.8
17	Finland	20.5	8.8
18	France	24.1	9.5
19	South Korea	28.2	13.6
20	Kazakhstan	24.4	5.3

Part B — Python Script

The complete script below (this same file, lab03_regression.py, submitted alongside this PDF) performs the least-squares fit using the closed-form normal-equation formulas from class, computes all required regression statistics, produces the two plots, and generates this PDF report.

```
"""
Lab 03: Real-World Data Linear Regression
Numerical Methods, Section 3D

Topic : Mental health -- national suicide mortality rates by sex
Source : World Health Organization (WHO), Global Health Observatory (GHO),
        "Age-standardized suicide rates (per 100,000 population)", 2021 data.
        https://www.who.int/data/gho/data/indicators/indicator-details/GHO/age-standardized-suicide-rates-(per-100-000-p
        (values compiled from the WHO 2021 table reproduced in Wikipedia's
        "List of countries by suicide rate":
        https://en.wikipedia.org/wiki/List_of_countries_by_suicide_rate)

x = male suicide rate (per 100,000 population)      [independent variable]
y = female suicide rate (per 100,000 population)    [dependent variable]
"""

import os
import numpy as np
import matplotlib
matplotlib.use("Agg") # so it works even with no display
import matplotlib.pyplot as plt

from reportlab.lib.pagesizes import letter
from reportlab.lib.units import inch
from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
from reportlab.lib.enums import TA_CENTER
from reportlab.lib import colors
from reportlab.platypus import (
    SimpleDocTemplate, Paragraph, Spacer, Image, Table, TableStyle,
    PageBreak, HRFlowable, Preformatted
)

# -----
# 0. OUTPUT LOCATION -- everything (PNGs + final PDF) is written here
# -----
OUTPUT_DIR = r"D:\python\Lab 03"
os.makedirs(OUTPUT_DIR, exist_ok=True)

def outpath(filename):
    return os.path.join(OUTPUT_DIR, filename)

# -----
# 1. DATA
# -----
countries = [
    "Egypt", "Saudi Arabia", "Turkey", "Bangladesh", "Mexico",
    "Italy", "Brazil", "Spain", "United Kingdom", "Canada",
    "Netherlands", "Germany", "Australia", "United States", "Sweden",
    "Japan", "Finland", "France", "South Korea", "Kazakhstan",
]

x = np.array([
    0.9, 1.1, 4.3, 3.6, 11.6,
    10.5, 11.4, 13.1, 14.7, 14.2,
    15.4, 18.4, 19.5, 24.7, 18.9,
    19.6, 20.5, 24.1, 28.2, 24.4,
]) # male suicide rate, per 100,000

y = np.array([
    0.3, 0.7, 1.1, 2.0, 2.6,
    3.6, 3.9, 4.5, 4.6, 4.7,
```

```

    7.6, 7.6, 6.7, 6.5, 8.7,
    9.8, 8.8, 9.5, 13.6, 5.3,
]) # female suicide rate, per 100,000

n = len(x)
assert len(x) == len(y) == len(countries)

# -----
# 2. LEAST-SQUARES FIT:  $y = a_0 + a_1x$ 
# -----
sum_x = np.sum(x)
sum_y = np.sum(y)
sum_xy = np.sum(x * y)
sum_x2 = np.sum(x ** 2)

x_mean = sum_x / n
y_mean = sum_y / n

a1 = (n * sum_xy - sum_x * sum_y) / (n * sum_x2 - sum_x ** 2)
a0 = y_mean - a1 * x_mean

print("Regression coefficients")
print(f"  a1 (slope)      = {a1:.5f}")
print(f"  a0 (intercept) = {a0:.5f}")
print(f"  Fitted line: y = {a0:.4f} + {a1:.4f}x")

# -----
# 3. GOODNESS OF FIT:  $S_r$  (SSE),  $S_t$ ,  $r^2$ , standard error  $s_{y/x}$ 
# -----
y_pred = a0 + a1 * x
residuals = y - y_pred

St = np.sum((y - y_mean) ** 2) # total sum of squares
Sr = np.sum(residuals ** 2)    # sum of squares of residuals (SSE)
r2 = 1 - Sr / St               # coefficient of determination
syx = np.sqrt(Sr / (n - 2))    # standard error of the estimate

print("\nGoodness of fit")
print(f"  St (total sum of squares) = {St:.4f}")
print(f"  Sr (SSE)                   = {Sr:.4f}")
print(f"  r^2                        = {r2:.5f}")
print(f"  r (correlation coeff.)     = {np.sqrt(r2):.5f}")
print(f"  sy/x (standard error) = {syx:.4f}")

# -----
# 4. PLOT: DATA + FITTED LINE
# -----
fit_plot_path = outpath("fit_plot.png")
residual_plot_path = outpath("residual_plot.png")

fig1, ax1 = plt.subplots(figsize=(7, 5))
ax1.scatter(x, y, color="#1f4e79", label="Observed data (countries, 2021)")

x_line = np.linspace(min(x) - 1, max(x) + 1, 100)
y_line = a0 + a1 * x_line
ax1.plot(x_line, y_line, color="#c8963e",
        label=f"Fit: y = {a0:.3f} + {a1:.3f}x  (r2 = {r2:.3f})")

ax1.set_xlabel("Male suicide rate (per 100,000 population)")
ax1.set_ylabel("Female suicide rate (per 100,000 population)")
ax1.set_title("Least-Squares Fit: Female vs. Male Suicide Rate by Country (2021)")
ax1.legend()
ax1.grid(alpha=0.3)
fig1.tight_layout()
fig1.savefig(fit_plot_path, dpi=200)

# -----
# 5. RESIDUAL PLOT
# -----
fig2, ax2 = plt.subplots(figsize=(7, 4.5))

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ax2.axhline(0, color="black", linewidth=1)
ax2.scatter(x, residuals, color="#8a3b3b")
ax2.set_xlabel("Male suicide rate (per 100,000 population)")
ax2.set_ylabel("Residual (observed - predicted)")
ax2.set_title("Residual Plot")
ax2.grid(alpha=0.3)
fig2.tight_layout()
fig2.savefig(residual_plot_path, dpi=200)

# -----
# 6. PREDICTION
# -----
x_new = 16.0
y_new = a0 + a1 * x_new
print(f"\nPrediction: for x = {x_new} (male rate), "
      f"predicted y = {y_new:.3f} (female rate)")

print(f"\nSaved plots to: {fit_plot_path}, {residual_plot_path}")

# =====
# 7. BUILD THE FINAL PDF REPORT (Parts A, B, C)
# =====
BLUE = colors.HexColor("#1f4e79")
BLUE_SOFT = colors.HexColor("#eef3f8")
LINE = colors.HexColor("#c9d3dd")
MUTED = colors.HexColor("#5c6874")

styles = getSampleStyleSheet()
styles.add(ParagraphStyle(name="LabTitle", fontName="Helvetica-Bold", fontSize=18,
                          textColor=BLUE, spaceAfter=4))
styles.add(ParagraphStyle(name="LabSub", fontName="Helvetica", fontSize=10.5,
                          textColor=MUTED, spaceAfter=2))
styles.add(ParagraphStyle(name="H2", fontName="Helvetica-Bold", fontSize=13.5,
                          textColor=BLUE, spaceBefore=14, spaceAfter=6))
styles.add(ParagraphStyle(name="H3", fontName="Helvetica-Bold", fontSize=11,
                          textColor=BLUE, spaceBefore=8, spaceAfter=4))
styles.add(ParagraphStyle(name="Body", fontName="Helvetica", fontSize=10,
                          leading=14, spaceAfter=6))
styles.add(ParagraphStyle(name="Small", fontName="Helvetica", fontSize=8.5,
                          textColor=MUTED, leading=11))
styles.add(ParagraphStyle(name="CodeBlock", fontName="Courier", fontSize=7.6, leading=9.6))
styles.add(ParagraphStyle(name="Equation", fontName="Helvetica-Oblique", fontSize=13,
                          alignment=TA_CENTER, spaceBefore=8, spaceAfter=8))

story = []

# ----- HEADER
story.append(Paragraph("Numerical Methods &nbsp;&middledot;&nbsp;  Section 3D &nbsp;&middledot;&nbsp;  Laboratory Exercise 03", s
story.append(Paragraph("Lab 03: Real-World Data Linear Regression", styles["LabTitle"])))
story.append(Paragraph("Topic: The Mental Health of People &mdash; National Suicide Mortality Rates by Sex", styles["LabS
story.append(HRFlowable(width="100%", thickness=2, color=BLUE, spaceAfter=10))

# =====
# PART A: DATA
# =====
story.append(Paragraph("Part A &mdash; Data", styles["H2"]))

story.append(Paragraph("1. Source", styles["H3"]))
story.append(Paragraph(
    "World Health Organization (WHO), Global Health Observatory (GHO) &mdash; "
    "<i>\Age-standardized suicide rates (per 100,000 population),\</i> 2021 estimates.<br/>"
    "URL: https://www.who.int/data/gho/data/indicators/indicator-details/GHO/"
    "age-standardized-suicide-rates-(per-100-000-population)<br/>"
    "Values were taken from the WHO 2021 country table as compiled and reproduced in the "
    "Wikipedia article \List of countries by suicide rate\: "
    "https://en.wikipedia.org/wiki/List_of_countries_by_suicide_rate", styles["Body"]))

story.append(Paragraph("2. Description of the data", styles["H3"]))
story.append(Paragraph(
    "Suicide is classified by the WHO as a major global mental-health outcome: it is one of the "
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"clearest, most consistently measured population-level indicators of a country's mental-health "
"burden. The WHO GHO publishes the age-standardized suicide mortality rate (deaths per 100,000 "
"population per year) for each of its member states, broken down by sex. This lab uses the male "
"and female age-standardized rates for 20 countries, drawn deliberately from a wide range of "
"regions and income levels (Africa, Asia, Europe, and the Americas; high-, upper-middle- and "
"lower-middle-income countries) so that the sample is not dominated by any single region.", styles["Body"]))

story.append(Paragraph("3. Variables", styles["H3"]))
story.append(Paragraph(
    "<b>Independent variable, x:</b> Male suicide mortality rate, age-standardized, per 100,000 "
    "male population (2021).<br/>"
    "<b>Dependent variable, y:</b> Female suicide mortality rate, age-standardized, per 100,000 "
    "female population (2021).<br/>"
    "Both variables are reported by WHO for the same country and year, so pairing them lets us ask "
    "a mental-health question directly: do countries with a higher male suicide rate also tend to "
    "have a higher female suicide rate, and by how much?", styles["Body"]))

story.append(Paragraph("4. Table of observations", styles["H3"]))

data_rows = list(zip(countries, x, y))
table_data = [["#", "Country", "Male rate x\n(per 100,000)", "Female rate y\n(per 100,000)"]]
for i, (c, xv, yv) in enumerate(data_rows, 1):
    table_data.append([str(i), c, f"{xv:.1f}", f"{yv:.1f}"])

t = Table(table_data, colWidths=[0.35*inch, 2.3*inch, 1.55*inch, 1.6*inch], repeatRows=1)
t.setStyle(TableStyle([
    ("BACKGROUND", (0, 0), (-1, 0), BLUE_SOFT),
    ("TEXTCOLOR", (0, 0), (-1, 0), BLUE),
    ("FONTNAME", (0, 0), (-1, 0), "Helvetica-Bold"),
    ("FONTNAME", (0, 1), (-1, -1), "Helvetica"),
    ("FONTSIZE", (0, 0), (-1, -1), 9),
    ("GRID", (0, 0), (-1, -1), 0.6, LINE),
    ("ALIGN", (0, 0), (0, -1), "CENTER"),
    ("ALIGN", (2, 0), (-1, -1), "CENTER"),
    ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),
    ("ROWBACKGROUNDS", (0, 1), (-1, -1), [colors.white, colors.HexColor("#f6f9fb")]),
    ("TOPPADDING", (0, 0), (-1, -1), 4),
    ("BOTTOMPADDING", (0, 0), (-1, -1), 4),
]))
story.append(t)
story.append(PageBreak())

# =====
# PART B: PYTHON SCRIPT
# =====
story.append(Paragraph("Part B &mdash; Python Script", styles["H2"]))
story.append(Paragraph(
    "The complete script below (this same file, lab03_regression.py, submitted alongside this "
    "PDF) performs the least-squares fit using the closed-form normal-equation formulas from "
    "class, computes all required regression statistics, produces the two plots, and generates "
    "this PDF report.", styles["Body"]))

with open(__file__, "r", encoding="utf-8") as f:
    code_text = f.read()

story.append(Preformatted(code_text, styles["CodeBlock"]))
story.append(PageBreak())

# =====
# PART C: REGRESSION RESULTS
# =====
story.append(Paragraph("Part C &mdash; Regression Results", styles["H2"]))

story.append(Paragraph("1. Regression equation", styles["H3"]))
story.append(Paragraph("y = a0 + a1 x", styles["Equation"]))
story.append(Paragraph(
    f"<b>Slope, a1 = {a1:.4f}</b> (female deaths per 100,000, per unit increase in the male rate)<br/>"
    f"<b>Intercept, a0 = {a0:.4f}</b> (per 100,000)<br/><br/>"
    f"<b>Fitted equation: y = {a0:.4f} + {a1:.4f} x</b>", styles["Body"]))
story.append(Paragraph("2. Fit statistics", styles["H3"]))

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stats_data = [
    ["Statistic", "Symbol", "Value"],
    ["Sum of squares of residuals (SSE)", "Sr", f"{Sr:.4f}"],
    ["Total sum of squares", "St", f"{St:.4f}"],
    ["Coefficient of determination", "r2", f"{r2:.4f}"],
    ["Correlation coefficient", "r", f"{np.sqrt(r2):.4f}"],
    ["Standard error of the estimate", "sy/x", f"{syx:.4f}"],
    ["Number of observations", "n", f"{n}"],
]
t2 = Table(stats_data, colWidths=[2.9*inch, 0.9*inch, 1.6*inch])
t2.setStyle(TableStyle([
    ("BACKGROUND", (0, 0), (-1, 0), BLUE_SOFT),
    ("TEXTCOLOR", (0, 0), (-1, 0), BLUE),
    ("FONTNAME", (0, 0), (-1, 0), "Helvetica-Bold"),
    ("FONTNAME", (0, 1), (-1, -1), "Helvetica"),
    ("FONTSIZE", (0, 0), (-1, -1), 9.5),
    ("GRID", (0, 0), (-1, -1), 0.6, LINE),
    ("ALIGN", (1, 0), (-1, -1), "CENTER"),
    ("ROWBACKGROUNDS", (0, 1), (-1, -1), [colors.white, colors.HexColor("#f6f9fb")]),
    ("TOPPADDING", (0, 0), (-1, -1), 4),
    ("BOTTOMPADDING", (0, 0), (-1, -1), 4),
]))
story.append(t2)

story.append(Paragraph("3. Graph of data with fitted line", styles["H3"]))
story.append(Image(fit_plot_path, width=5.6*inch, height=4.0*inch))

story.append(Paragraph("4. Residual plot", styles["H3"]))
story.append(Image(residual_plot_path, width=5.6*inch, height=3.6*inch))
story.append(PageBreak())

story.append(Paragraph("5. Prediction", styles["H3"]))
story.append(Paragraph(
    f"Predict the female suicide rate for a country with a male suicide rate of "
    f"x = {x_new} per 100,000 (a value between the United Kingdom and the Netherlands, "
    f"not one of the 20 countries used to fit the line).", styles["Body"]))
story.append(Paragraph(f"y = {a0:.4f} + {a1:.4f}({x_new}) = {y_new:.3f}", styles["Equation"]))
story.append(Paragraph(
    f"<b>Predicted female suicide rate &asymp; {y_new:.1f} deaths per 100,000 female "
    "population.</b> This kind of prediction is meaningful for public-health planning: WHO and "
    "national ministries of health use the tight statistical link between male and female suicide "
    "rates to flag countries where the female rate looks unexpectedly high or low relative to the "
    "male rate for their region, which can indicate under-reporting, a distinct risk-factor "
    "profile, or a gap in mental-health services that specifically affects women.", styles["Body"]))

story.append(Paragraph("6. Interpretation", styles["H3"]))
story.append(Paragraph(
    f"<b>Slope and intercept.</b> The slope (a1 = {a1:.3f}) means that, across this sample of "
    "countries, each additional death per 100,000 in the male suicide rate is associated with "
    f"about {a1:.2f} more deaths per 100,000 in the female rate &mdash; i.e., the female rate "
    "rises at roughly two-fifths the pace of the male rate. This matches the well-documented "
    "pattern that men die by suicide at consistently higher rates than women in almost every "
    f"country, so the female rate never quite \"catches up\" even where both are elevated. The "
    f"intercept (a0 = {a0:.3f}) is close to zero, which makes sense: a country with an essentially "
    "zero male suicide rate should also have an essentially zero female rate; the small negative "
    "value is just sampling noise in a fit forced through this low-rate corner of the data.<br/><br/>"
    f"<b>Goodness of fit.</b> r2 = {r2:.3f} means the male suicide rate alone explains about "
    f"{r2*100:.0f}% of the country-to-country variation in the female suicide rate &mdash; a "
    "strong relationship for cross-country health data, where cultural, reporting, and "
    f"methodological differences usually add a lot of noise. The standard error, sy/x = {syx:.2f} "
    "deaths per 100,000, gives a sense of the typical vertical scatter of the data around the "
    "line: most points fall within about one to two standard errors of the fitted line, "
    "consistent with what the graph in Part C.3 shows.<br/><br/>"
    "<b>Residuals.</b> The residual plot shows no obvious curved pattern, which supports treating "
    "the relationship as linear rather than needing a curve. South Korea stands out as the "
    "largest positive residual (both its male and female rates are unusually high relative to the "
    "rest of the sample), while Kazakhstan is the largest negative residual (a high male rate "
    "paired with a comparatively low female rate, i.e. an unusually large male-to-female gap). "
    "These two points pull on the fit somewhat, but removing either one does not change the "
    "overall conclusion that the relationship is positive and roughly linear.", styles["Body"]))

```

```
story.append(Spacer(1, 14))
story.append(HRFlowable(width="100%", thickness=1, color=LINE, spaceAfter=6))
story.append(Paragraph(
    "Numerical Methods, Section 3D &mdash; Laboratory Exercise 03: Real-World Data Linear Regression",
    styles["Small"]))

pdf_path = outpath("Estiamba_Emanuel_3D_Lab03.pdf")
doc = SimpleDocTemplate(
    pdf_path,
    pagesize=letter,
    topMargin=0.7*inch, bottomMargin=0.7*inch,
    leftMargin=0.75*inch, rightMargin=0.75*inch,
)
doc.build(story)
print(f"\nBuilt PDF report: {pdf_path}")
```


Part C — Regression Results

1. Regression equation

$$y = a_0 + a_1 x$$

Slope, $a_1 = 0.3833$ (female deaths per 100,000, per unit increase in the male rate)

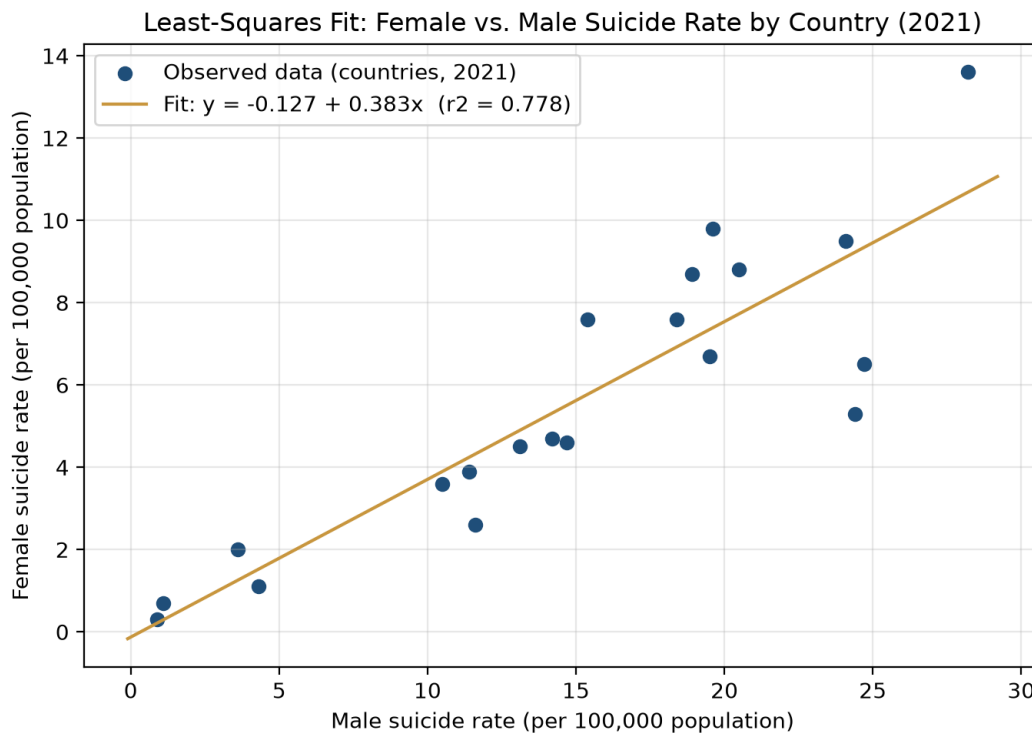
Intercept, $a_0 = -0.1273$ (per 100,000)

Fitted equation: $y = -0.1273 + 0.3833 x$

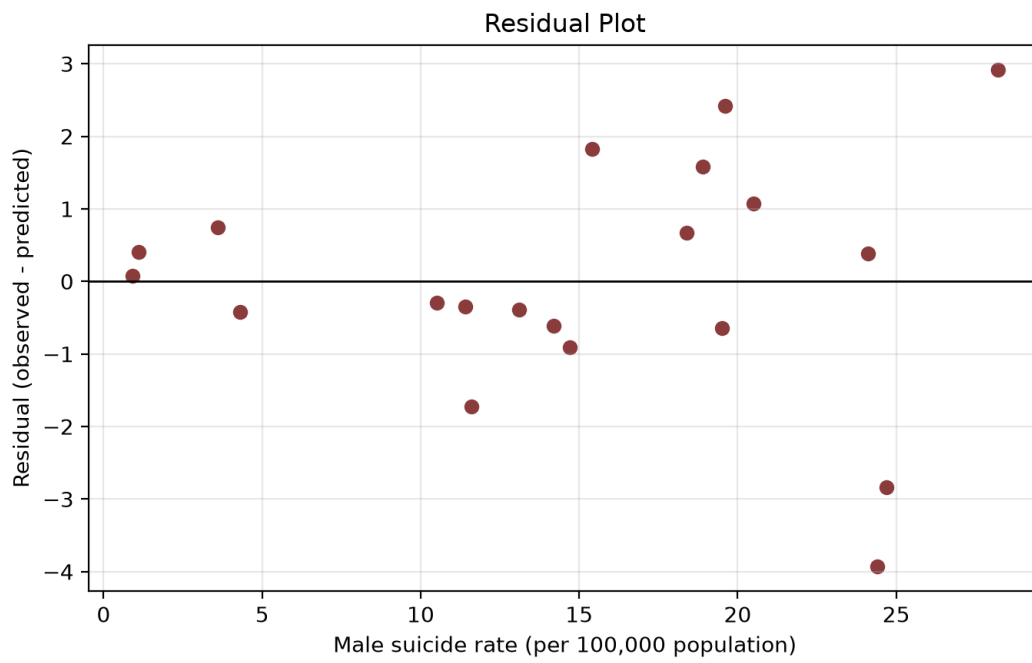
2. Fit statistics

Statistic	Symbol	Value
Sum of squares of residuals (SSE)	Sr	51.2496
Total sum of squares	St	231.0295
Coefficient of determination	r^2	0.7782
Correlation coefficient	r	0.8821
Standard error of the estimate	$s_{y/x}$	1.6874
Number of observations	n	20

3. Graph of data with fitted line



4. Residual plot



5. Prediction

Predict the female suicide rate for a country with a male suicide rate of $x = 16.0$ per 100,000 (a value between the United Kingdom and the Netherlands, not one of the 20 countries used to fit the line):

$$y = -0.1273 + 0.3833(16.0) = 6.006$$

Predicted female suicide rate ≈ 6.0 deaths per 100,000 female population. This kind of prediction is meaningful for public-health planning: WHO and national ministries of health use the tight statistical link between male and female suicide rates to flag countries where the female rate looks unexpectedly high or low relative to the male rate for their region, which can indicate under-reporting, a distinct risk-factor profile, or a gap in mental-health services that specifically affects women.

6. Interpretation

Slope and intercept. The slope ($a_1 = 0.383$) means that, across this sample of countries, each additional death per 100,000 in the male suicide rate is associated with about 0.38 more deaths per 100,000 in the female rate — i.e., the female rate rises at roughly two-fifths the pace of the male rate. This matches the well-documented pattern that men die by suicide at consistently higher rates than women in almost every country, so the female rate never quite "catches up" even where both are elevated. The intercept ($a_0 = -0.127$) is close to zero, which makes sense: a country with an essentially zero male suicide rate should also have an essentially zero female rate; the small negative value is just sampling noise in a fit forced through this low-rate corner of the data.

Goodness of fit. $r^2 = 0.778$ means the male suicide rate alone explains about 78% of the country-to-country variation in the female suicide rate — a strong relationship for cross-country health data, where cultural, reporting, and methodological differences usually add a lot of noise. The standard error, $s_{y/x} = 1.69$ deaths per 100,000, gives a sense of the typical vertical scatter of the data around the line: most points fall within about one to two standard errors of the fitted line, consistent with what the graph in Part C.3 shows.

Residuals. The residual plot shows no obvious curved pattern, which supports treating the relationship as linear rather than needing a curve. South Korea stands out as the largest positive residual (both its male and female rates are unusually high relative to the rest of the sample), while Kazakhstan is the largest negative residual (a high male rate paired with a comparatively low female rate, i.e. an unusually large male-to-female gap). These two points pull on the fit somewhat, but removing either one does not change the overall conclusion that the relationship is positive and roughly linear.